

**BAYESIAN DATA ANALYSIS
ASSIGNMENT 1**

Due: Monday, 2 October 2023

PROBLEM 1

X_1 and X_2 have joint PMF

x_1	x_2	$\text{Prob}(X_1 = x_1, X_2 = x_2)$
0	0	0.15
1	0	0.15
2	0	0.15
0	1	0.15
1	1	0.20
2	1	0.20

- Compute the marginal distribution of X_1
- Compute the marginal distribution of X_2
- Compute the conditional distribution of $X_1|X_2$
- Compute the conditional distribution of $X_2|X_1$
- Are X_1 and X_2 independent? Justify your answer.

PROBLEM 2

According to insurance.com, the 2017 auto theft rate was 135 per 10,000 residents in Raleigh NC compared to 214 per 10,000 residents in Durham/Chapel Hill. Assuming Raleigh's population is twice as large as Durham/Chapel Hill and a car has been stolen somewhere in one of these two areas, what the probability it was stolen in Raleigh?

PROBLEM 3

Your daily commute is distributed uniformly between 15 and 20 minutes if there is no convention downtown. However, conventions are scheduled for roughly 1 in 4 days, and your commute time is distributed uniformly from 15 to 30 minutes if there is a convention. Let Y be your commute time this morning.

- What is the probability that there was a convention downtown given $Y = 18$?
- What is the probability that there was a convention downtown given $Y = 28$?